

Experimental Results when Playing Chromatically on a Diatonic Harmonica

L. Millot, Ch. Cuesta, C. Valette
Laboratoire d'Acoustique Musicale
(CNRS; Université Paris 6; Ministère de la Culture) UPMC
- case 161, 4 place Jussieu, F-75252 Paris cedex 05, France.

Abstract

One can play chromatically on a diatonic harmonica by using different playing modes: normal playing, bends and overnotes (overblows and overdraws). For each playing mode, the reed motions were measured with strain gauges and the inner pressures were recorded at the same time with a differential pressure transducer. The reed motions were shown to be sinusoidal. The control of active reed motion(s) on the evolutions of the inner pressure was underlined. The internal mechanisms occurring during playing were systematically analyzed. Modelling of the reeds operation was derived and explanations for the harmonic content of the sound is given.

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1 Introduction

Much attention has been paid to the study of reed instruments like the saxophone or clarinet. The interested reader may begin to search some information in [1] or [2]. The present article deals with a physical instrument poorly studied : the diatonic harmonica. The main difference between instruments like the clarinet or saxophone and the diatonic harmonica is that the respective natures of the reeds are quite different. For the saxophone or the clarinet, the reed is a weak (operating far under its first natural frequency) and beating (a significant closing period is common in normal playing) reed [3] : we refer in this article to this kind of reed as "striking reed". Concerning the diatonic harmonica, the reed, made of smooth alloy, is a strong (operating around its first natural frequency) and free (the reed is able to go through its support during normal playing) reed [4] : we refer to this kind of reed as "free reed".

Some harmonicists, such as H. Lewy or J. Hanriot, are able to play fully chromatically (at least) on a diatonic harmonica [5, 6] using three kinds of different playing modes: normal blown and drawn playing modes, blown and drawn bends, overnotes (overblows and overdraws). But, for the moment, each player needs very long practice and empirical search before being able to do it. Two experimental studies [7, 8] underline the existence of a role of the oral cavity in the playing of the harmonica. On the other hand, one mainly finds qualitative experimental results about the reed motions during the playing of each mode [9, 8, 10]. The present work is a part of a study of the diatonic harmonica, initiated at the Laboratoire d'Acoustique Musicale by the questions of J. Hanriot, professional harmonicist, on chromatically playing the diatonic harmonica.

In this paper, we give some quantitative experimental results about the phenomena at work inside the harmonica when playing every possible mode. As the radiated sound is only a negligible part of the running phenomena (attenuation of 80 dB of the sound pressure outside with respect to the inner one), we only measured the inner pressure difference between the upstream and downstream of the reeds. Instead of making use of some proximity displacement sensors as Bahnson did [8], because of their non linearity, we preferred to use strain gauges stuck on the reeds, even if this modified the tuning of the reeds. But we only kept for measurements reeds for which the quality of tone was acceptable compared to the usual one for a non-equipped harmonica.

First, with these experiments we made clear the fact that, while playing, both free reed motions were sinusoidal for the diatonic harmonica, whatever playing mode was used: we got sinusoidal oscillations around a displaced equilibrium position. Second, we verified the frequency conditions and the vibrational behaviour assumed for each reed: active or passive reeds, pitch of the played note relative to the plucked natural frequency of the reed, *inward striking* or *outward striking* character of the reed [11, 12, 13]. Third, we also elucidated the main control of active reed(s) on the variations of the inner pressure. With these experimental results, we were able to model each free reed using an equivalent one-degree of freedom damped oscillator (mass-stiffness-damper system) whatever playing mode was studied. As the reed motions were sinusoidal, by studying the inner pressure evolutions, we also think that the only non-linear part of the system, responsible for the very rich timbre of the harmonica notes, is the air flow through the reeds.

2 Experimental device

2.1 Device

For convenience we used the experimental design shown on Figure 1, made up of the following parts: an harmonica, two pairs of strain gauges and the two related half Wheatstone bridges, a differential pressure transducer and its dedicated electronics, and a data acquisition system installed in a computer. The location of the outer microphone used to measure the external over-pressure is also indicated on Figure 1.

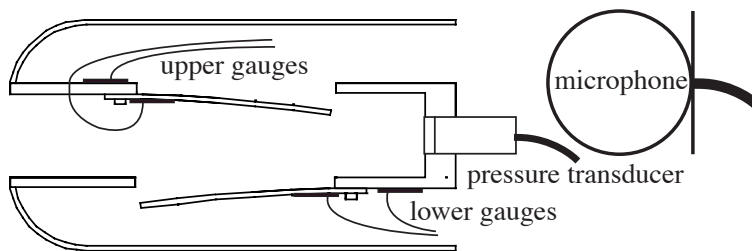


Figure 1: Scheme of the experimental device used for the measurement of the reed motions and for the recording of the inner and outer over-pressures.

For the harmonica, we used Lee Oskar TM diatonic models because of their modular design which allows easy assembly and disassembly. We chose the G tonality because the related reeds are the biggest ones and then should be less sensitive to the attachment of gauges. Furthermore,

with the G tonality (the lower one), the tested frequencies are the lowest and we should be in the optimal running area for all the measurements devices (linear range and minimal dephasing). Indeed, using the lowest frequencies should improve the agreement between the experimental results and the ones calculated with a one-dimensional modelling. We decided to fit the channels 4 and 7 out with strain gauges because, with these two channels, one can produce playing modes of each kind much more easily. It was very important to choose channels for which the overnotes are the easiest because, when the reeds are fitted out with gauges, these overnotes are very difficult or even impossible to produce. We fitted two harmonicas on the fourth channel (referred as harmonicas 4a and 4b) and two other harmonicas on the seventh channel (referred as harmonicas 7a and 7b).

For each of the fitted reeds, we made the SEEA company cold weld an active strain gauge on the reed as near as possible to the embedding and a passive one on the reed support. A half Wheastone bridge was achieved by this pair of gauges. The strain gauges used were PA XX 050 BA 120 models (120 Ω gauges with a barylen shield layer). As these gauges proved to be too fragile, two new harmonicas are to being fitted with hot welded Vishay Measurements gauges: four tiny gauges on each reed, making a complete Wheastone bridge.

Because of the tiny dimensions of each channel we used a 8507 C-2 Endevco differential pressure transducer (piezoresistive pressure transducer featuring an active four-arm strain gauge bridge diffused into a sculptured silicon diaphragm) to measure the pressure difference between upstream and downstream of the reeds (inner pressure). The outer diameter of the sensor is 2.34 mm while the channel section is a square of 5 mm side. With this sensor we have a over-pressure scale from zero to 14 kPa, a upper linear limit frequency of 14 kHz, a 23.32 $\mu\text{V}.\text{Pa}^{-1}$ sensitivity under 10 V stabilised alimentation, a full scale output signal of 322 mV for added errors (non-linearity, hysteresis and non repeatability) equal to 0.82% of the full scale output.

The data acquisition system was made up with a PCI-MIO-16E-4 National Instrument card, a BNC-2090 National Instrument extension kit and LabView 5.01 installed on a Pentium 200 MMX. With this system, we were able to record our three signals (reeds openings and inner pressure) with a 40 kHz sampling frequency and a 12 bits dynamic.

2.2 Calibration

All the fitted reeds were calibrated using a customized Mettler electronic balance (plate replaced by a sharp conical needle used to push the reed tip) which allowed us to measure the tip reed displacement, the disequilibrium voltage of the Wheatstone brige and the force at the tip of the reed simultaneously. Figure 2 gives a typical example of the calibration measurements: a perfect linearity, on the first hand, between the force and the tip reed displacement and, on the other hand, between the disequilibrium voltage and the tip reed displacement.

As suspected, sticking on a gauge increased the plucked natural frequency of the reed, as shown by Table 1 ¹. Indeed, if the stiffness increased very much with the gauge sticking, the mass did not increase as much as the stiffness does. This increase of the natural frequency changed the

¹There are no calibration results for harmonica 7a because the both gauges died before this operation, during the first attempts to blow or draw on harmonica 7b. We must note that other gauges died during the experiments, and above all during the attempts to produce the overnotes.

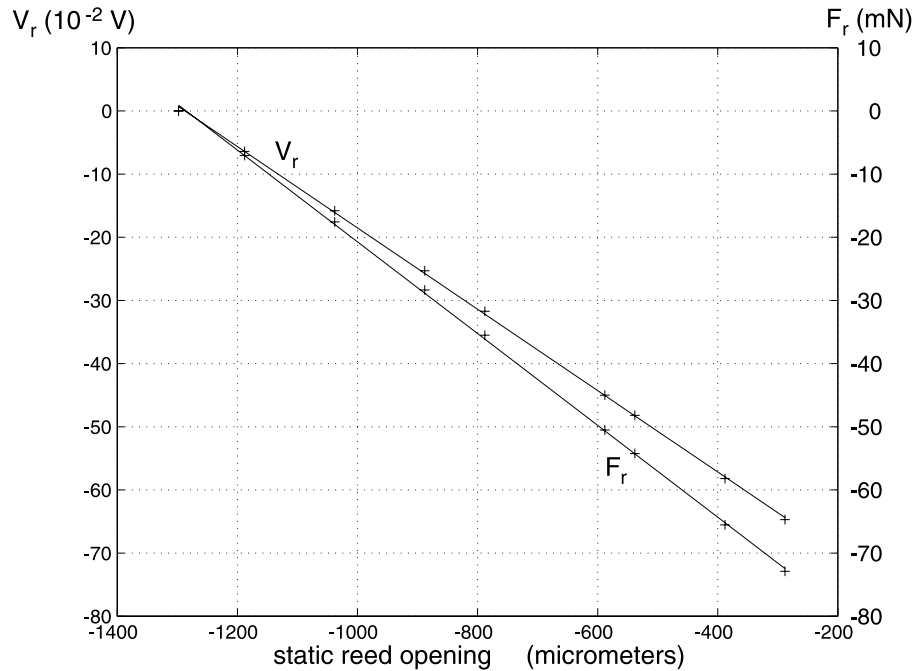


Figure 2: Results for the calibration of the upper 4b reed: return force applied on the reed (measurements: $\times$; linear regression: $-$) and disequilibrium voltage (measurements: $+$; linear regression: $-$) function of the tip reed opening.

control of the player on the production of the notes. As a consequence, the abnormal playing modes (bends and overnotes) became hard or even impossible to produce: the control acquired by the player did not match that now needed; the player needed to learn and find the new control before being able to try to play the bends and above all the overnotes.

Table 1: Theoretical natural frequencies and effective ones after gauges implantation.

channel	reed	harmonica	freq. (Hz)	diff. (cents)
4	upper	theory	392	
		4a	398	+1.53
		4b	405	+3.32
4	lower	theory	442	
		4a	469	+6.11
		4b	455	+2.94
7	upper	theory	784	
		7b	807	+2.93
7	lower	theory	734	
		7b	775	+5.58

We decided that the reeds were not retuned and their rest openings (offsets) were not adjusted because the gauges were too fragile : there were too many risks of destroying the gauges while performing this kind of operation.

3 Study of the reeds

3.1 Reed motion

In the following section, we use the active and passive reed terms during the study of each playing mode. According to us, an active reed is a reed absolutely necessary to produce the note while, if a passive reed is kept blocked, the note can still be produced with the same fundamental but, sometimes, with a different tone.

For all the experiments, we noted that the reed motion was composed of sinusoidal oscillations around a displaced equilibrium position. Indeed, as illustrated by Table 2 and Figure 3, the rate of non-linear distortion for each reed was always less than 3.14% and there was almost only the fundamental mode present in the power spectrum of the reed movements. In fact, for each kind of mode, the bigger rate of non-linear distortion corresponds to the passive reeds when playing and the lower one to the active reed, the one responsible for the sound. These results are quite different from the ones found by Bahnson, Antaky and Beery [8] but we do believe that they used sensors with non-linear behaviour (they do not present a linearity test in their paper). These results are, of course, far different too from the ones established for the striking reed in the clarinet or in the saxophone [3], but this should not be surprising because the harmonica makes use of free reeds only.

Table 2: Rates of non-linear distortion for all the reeds when playing (when available).

	reed	blow	draw	bend	overnote
upper 4		0.45%	0.47%	0.08%	—
lower 4		3.14%	0.98%	0.35%	2.01%
upper 7		1.06%	1.19%	2.69%	2.81%
lower 7		0.20%	0.27%	1.53%	—

For free reeds, there is no obstacle to stop the reed motion, and it is not incomprehensible to find that the reed has a sinusoidal motion. We think that the free reed, in the harmonica, is a self-sustained oscillator operating at a frequency around its first transverse vibration eigen frequency. As in the harmonica the reeds are made of smooth alloy and the player’s blowing or drawing power is not very important (compared with that possible with an artificial blowing or drawing device) we think that one should hardly encounter a torsion vibrational mode during normal playing.

3.2 Equivalent oscillator

As the reeds were operating at a frequency around their first transverse vibration eigenfrequency, we thought that we could introduce a damped oscillator with one degree of freedom (mass-spring-damping) as an equivalent for each reed. As the rest opening (or offset) h_{n00} , the quality factor Q_n and the natural frequency f_{rn} for the reed n were given by the calibration, we just needed to define how to calculate the other characteristics of the equivalent oscillator: the stiffness K_n , the mass M_n and the damping R_n .

First, with the help of the reed calibration we calculated a tip reed stiffness $K_{cal,n}$ for the reed n , by using the static displacement for a beam with a pinpoint loading on the free tip

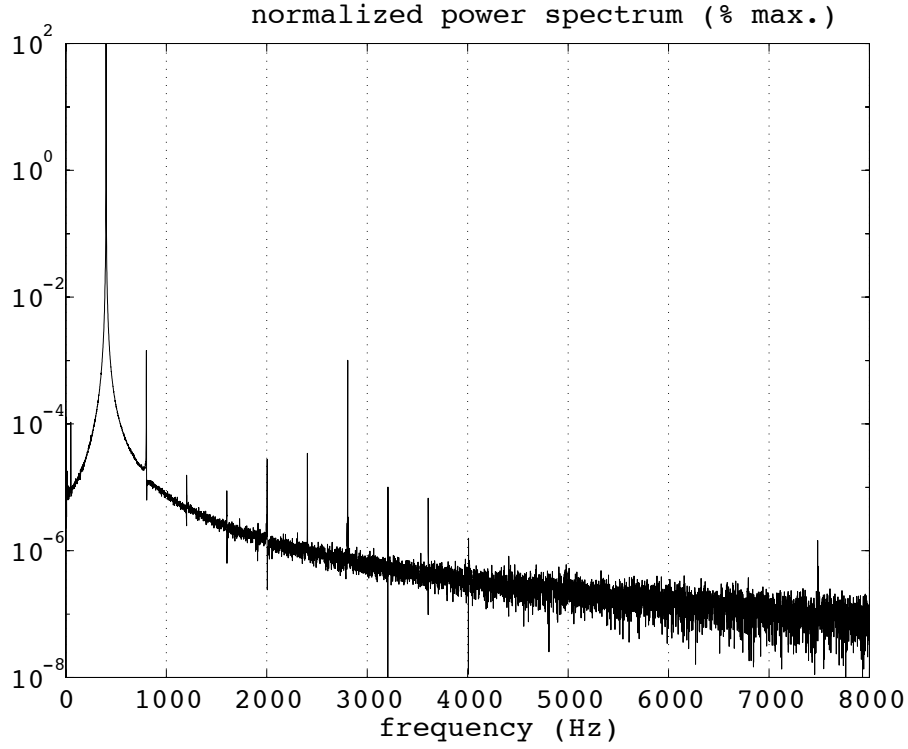


Figure 3: Normalized power spectrum of the upper reed motion for a normal blow on channel 4.

[14, 3, 4]. Second, we supposed that our reed-equivalent stiffness K_n should be calculated from the static displacement in the case when we assumed an uniform loading of the reed. Thus, with the expressions for both stiffnesses we were able to derive the expression linking $K_{cal,n}$ and K_n . The equivalent mass and damping were then determined with the quality factor Q_n , the eigen frequency f_{rn} and the equivalent stiffness K_n by a classical method. So, we have the following relationships which define the equivalent oscillator:

- $K_n = \frac{16}{15} K_{cal,n}$
- $M_n = \frac{K_n}{(2\pi f_{rn})^2}$
- $R_n = \frac{K_n}{2\pi f_{rn}} Q_n^{-1}$

With the calibration results we derived the characteristics of the equivalent oscillators for each fitted and tested reed, given by Table 3. In Table 3, upper and lower refer respectively to the upper reed and the lower one; 4a, 4b and 7b refer to harmonica 4a, harmonica 4b and harmonica 7b. The convention for the reed openings, similar to the one used by Fletcher [11] for the opening reed, is given by figure 4. These characteristics would be helpful for numerical simulations based on a one-dimensional model [4].

In the next step we studied what happened in the case of each playing mode for the reeds and the inner pressure.

Table 3: Mechanical characteristics for the six equivalent oscillators : plucked natural frequency f_{rn} , stiffness K_n , quality factor Q_n , mass M_n , damping R_n and rest opening h_{n00} .

reed	f_{rn} (Hz)	K_n (N/m)	Q_n	M_n (mg)	R_n ($\mu\text{N.s/m}$)	h_{n00} (μm)
upper 4a	398	73.5	246	11.75	119	-1343.2
lower 4a	469	48.6	110	5.60	150	+292.6
upper 4b	405	77.5	177	11.98	172	-1298.0
lower 4b	455	49.6	95	6.07	183	+286.5
upper 7b	807	91.8	228	3.57	79	-995.0
lower 7b	775	69.5	117	2.93	122	+166.4

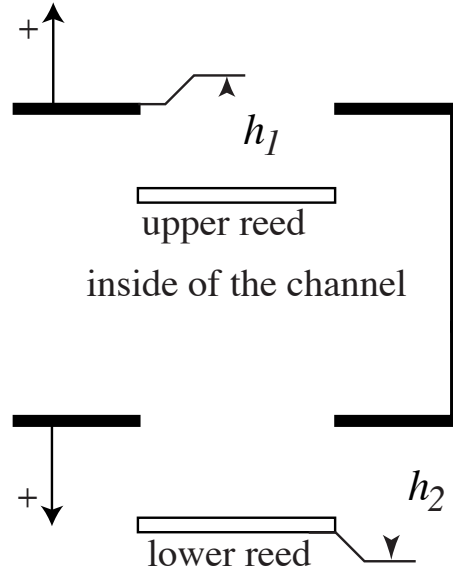


Figure 4: Illustration of the convention adopted for the reed openings.

4 Study of the three types of modes

To describe the three types of playing modes that exist, we give the experimental results for a normal blow on channel 4 with the harmonica 4b, a drawn bend on channel 4 with the harmonica 4ba² and an overdraw on channel 7 with the harmonica 7b. To calculate the pressure levels in dB, we assumed that 0 dB level corresponds to a reference pressure of 20 μPa . For the three described playing modes, the power spectrum was normalized to the maximum power: each power value was divided by the maximum power and given in percents.

In order to evaluate the *inward* or *outward striking* character of both reeds we introduced the quantity $\Delta h_{n0} = h_n - h_{n00}$, corresponding to the difference between the mean opening during playing and the unblown opening (offset at rest). For the upper reed, the reed inside of the channel, a positive Δh_{n0} indicates that the reed is an closing reed while a negative Δh_{n0} indicates an opening reed. For the lower reed, located outside of the channel, we have an opening

²We rebuilt an experimental harmonica with upper reed from harmonica 4b and lower reed from harmonica 4a after the death of two gauges: the upper on harmonica 4a and the lower on harmonica 4a.

reed if Δh_{n0} is positive and a closing reed if Δh_{n0} is negative.

4.1 Normal blow on channel 4

All the experimental results for both reeds are given by Figure 5 and Table 4. One can note that both reeds vibrate at the same frequency (401 Hz) and that the played frequency is less than the upper reed eigen frequency (405 Hz), as predicted for an upstream-striking reed [13].

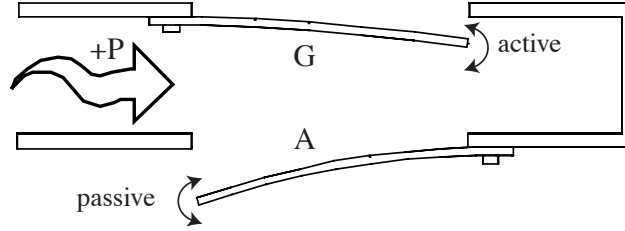


Figure 5: Normal blow configuration according to theory for a blow on channel 4.

Table 4: Results during a blow on channel 4; harmonica 4b.

reed	f_{play} (Hz)	H_n (μm)	Δh_{n0} (μm)	h_{n0} (μm)
upper 4b	401	1493	57	-1171
lower 4b	401	67	34	+320

As the acoustical amplitude of vibration for the upper reed, H_1 , is much bigger than H_2 , the active reed (the one driving the note frequency) is the upper reed while the lower reed is a passively vibrating reed, as illustrated on Figure 5: the upper reed auto-oscillates while the lower reed is forced to vibrate. One can also note that the acoustical amplitude H_1 of the upper reed oscillation is very important (1.493 mm) compared with the offset h_{100} (-1298 μm) and the equilibrium opening h_{10} (-1171 μm). In such a configuration, the upper reed goes through its support: free reeds might go through their support when the amplitude is high enough while striking reeds are stopped by the shallot (clarinet or saxophone reeds for instance). Moreover, this result clarifies that, even if the amplitude of the reed displacement is quite important, the reed motion stays sinusoidal, which is not the case for striking reeds. At last, studying the displacement of the mean opening for both reeds, given by the quantity Δh_{n0} , one can note that the expected classical result [11, 12, 13] for both reeds is verified: the upper reed is a closing reed when blown while the lower reed is an opening one.

The analysis of the inner pressure shows that the mean over-pressure is 219 Pa while the equivalent acoustical amplitude is 128 Pa. So, for a normal blow, one need to develop a weak over-pressure in order to produce the note. In the same kind of idea, the mean and maximum pressure levels are respectively 104.8 dB and 144.8 dB. With Figure 6 giving the normalized power spectrum for the inner over-pressure, we note that the fundamental is the most powerfull harmonic. The results show that the sixth and seventh harmonics, frequencies around the second transversal vibration mode of the beam, do not have a lot of power, which will need to be explained. Moreover, the odd harmonics have more power than the following even ones.

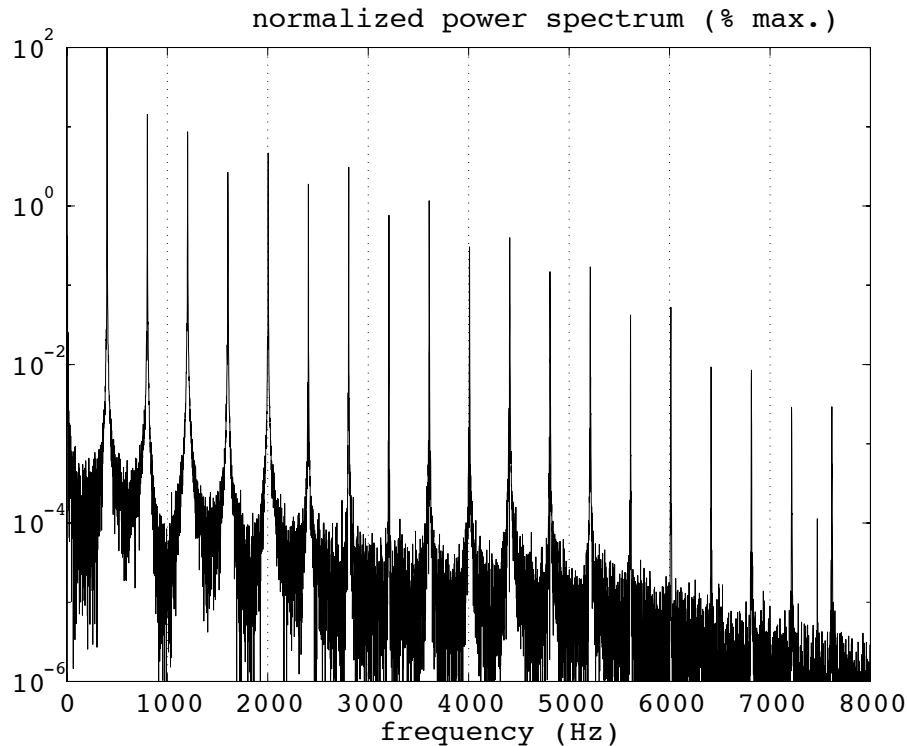


Figure 6: Normalized power spectrum for the internal over-pressure during a blow on channel 4.

Using Figure 7, the study of the reed motions' interaction and of their influence on the inner pressure can be done. One can point out the main influence, in the signal of the inner over-pressure, of the upper reed motion and the very small influence of the lower reed motion. Indeed, the lower reed opening is not powerful enough to perturb the upper reed influence during most of the period of the waveform. In the inner over-pressure waveform, several stages can be distinguished:

- a violent decrease of the inner over-pressure when the upper reed increases outside the channel during stage a (the reed has gone through the support !);
- an increase of the over-pressure, during stage b, when the upper and the lower reeds join their maximum opening;
- during stage c, the over-pressure begins to decrease with both reed openings;
- a severe decrease of the over-pressure occurs when the upper reed has gone back inside the channel, going through the support (stage d), the minimal over-pressure being related to the maximal opening of the upper reed inside of the channel ;
- during stage e, the over-pressure increases while the upper reed tends to close, with a violent rise of the over-pressure (peak) just before the upper reed closes the opening : as the lower reed opening increases just as this moment the over-pressure peak is not related to the closing of the upper reed;
- a new decrease of the over-pressure level which initiates a new cycle for the three waveforms.

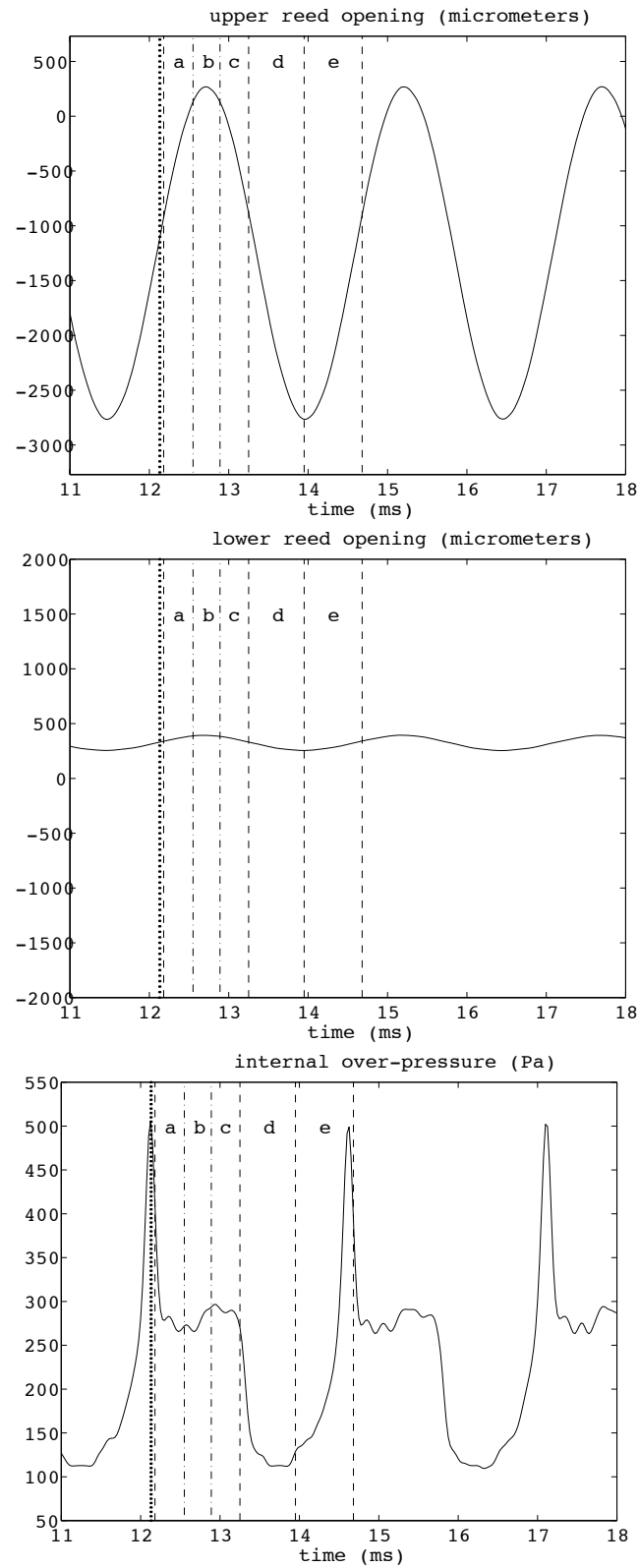


Figure 7: Signals for the upper reed motion, lower reed motion and internal over-pressure during a normal blow on channel 4. The dotted line indicates the over-pressure peak, the dashed lines the inside closing point of the upper reed and the dashdot lines its outside closing point.

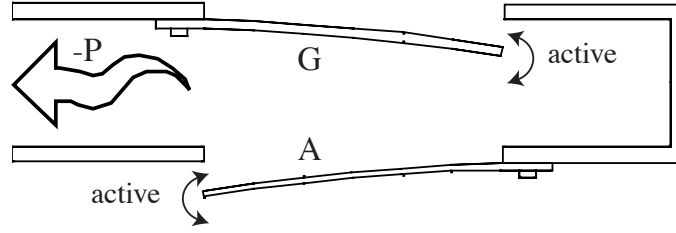


Figure 8: Bend configuration according to theory for a drawn bend on channel 4.

The sharp peak in the over-pressure wave form, associated with the crossing of the upper reed (return cross), is responsible for the richness of the sound, illustrated by the high number of harmonics in the over-pressure power spectrum (Figure 6). So, even with sinusoidal motions of the reed, the resulting acoustical field exhibits a very rich harmonic content because of the non-linearity of the flow through the valve, regulated by the reed opening. We must point out that the upper reed opening associated with the over-pressure peak is not associated with the inside closing point of the upper reed. Indeed, at this time, the lower reed opening increases and is responsible of the drop of the over-pressure : this is the only moment when the lower reed influence is significant.

4.2 Draw bend on channel 4

As for the normal blow on channel 4, we give all the experimental results for the reeds in the case of a drawn bend on channel 4 on harmonica 4ba (Figure 8 and Table 5).

Table 5: Results during a drawn bend on channel 4; harmonica 4ba.

reed	f_{play} (Hz)	H_n (μm)	Δh_{n0} (μm)	h_{n0} (μm)
upper 4b	433	409	-114	-1413
lower 4a	433	762	-200	+93

The main experimental results for both reeds are the following:

- the note is produced by the simultaneous auto-oscillation of both reeds (amplitude of 409 μm for the upper reed and of 762 μm for the lower reed), which means that both reeds participate to the note production;
- the amplitude of motion for the lower reed is higher than the one for the upper reed: this is because we have a drawn bend of a half-tone;
- the note pitch is between both eigen frequencies and is almost associated with a half-tone alteration of the drawn note (433 Hz instead of 431 Hz for the perfect half-tone, because of the gauge sticking);
- $\Delta h_{10} < 0$ so the upper reed is an opening reed while the lower reed is a closing reed, as we have $\Delta h_{20} < 0$;
- the upper reed runs over its eigen frequency: it is a so called *downstream striking reed* [11, 12, 13];

- the lower reed runs under its eigen frequency: it is a so called *upstream striking reed*.

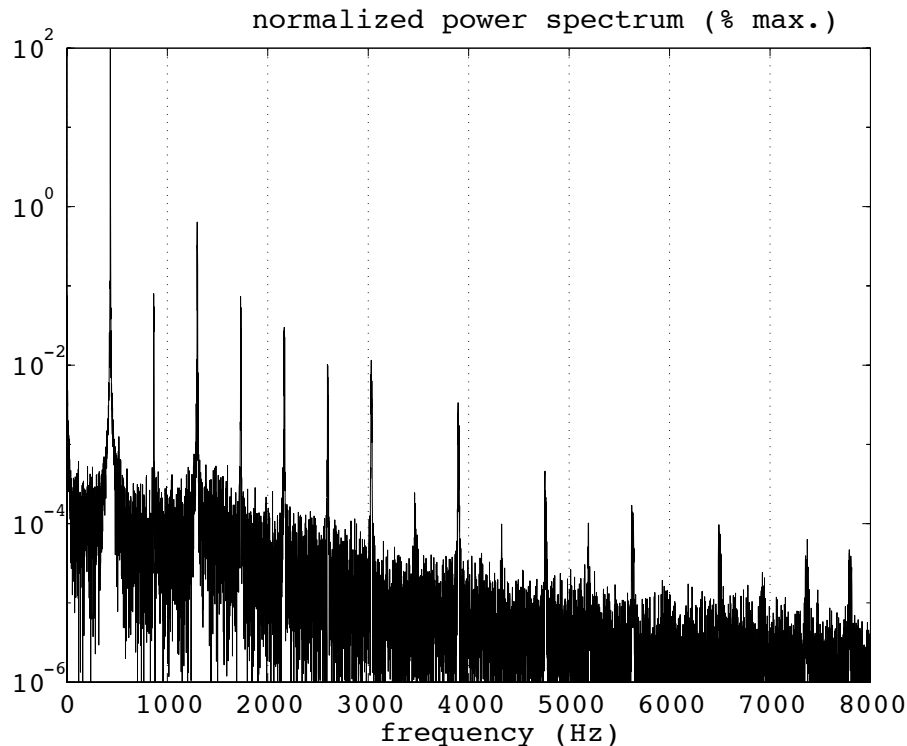


Figure 9: Normalized power spectrum for the internal pressure during a drawn bend on channel 4.

With regard to the inner pressure, we have a mean depression (drawn bend) of 635 Pa and acoustical variations of 397 Pa around this mean depression. The related mean and maximal pressure levels are 150 dB and 154.2 dB. Note that the supply depression needed to produce the drawn bend is higher than the supply over-pressure necessary for a normal blow (219 Pa), or the supply depression for a normal draw (230 Pa). This is one of the possible clues to explain why it is more difficult to produce bends rather than normal notes. The study of the power spectrum for the inner depression shows that:

- the fundamental is the highest level harmonic (0 dB level), as the following important harmonic has a level of -22 dB;
- unlike the normal modes the third harmonic is more powerful than the second one in the case of the drawn bend;
- the powers of the harmonics for the drawn bend, fundamental excepted, are weaker than the power of the harmonics for the normal mode, which might be explained by the fact that it is more difficult to produce a bend;
- the odd harmonics are more powerful than the following even ones as for the normal blown note;
- the sixth and seventh harmonics do not present a lot of power.

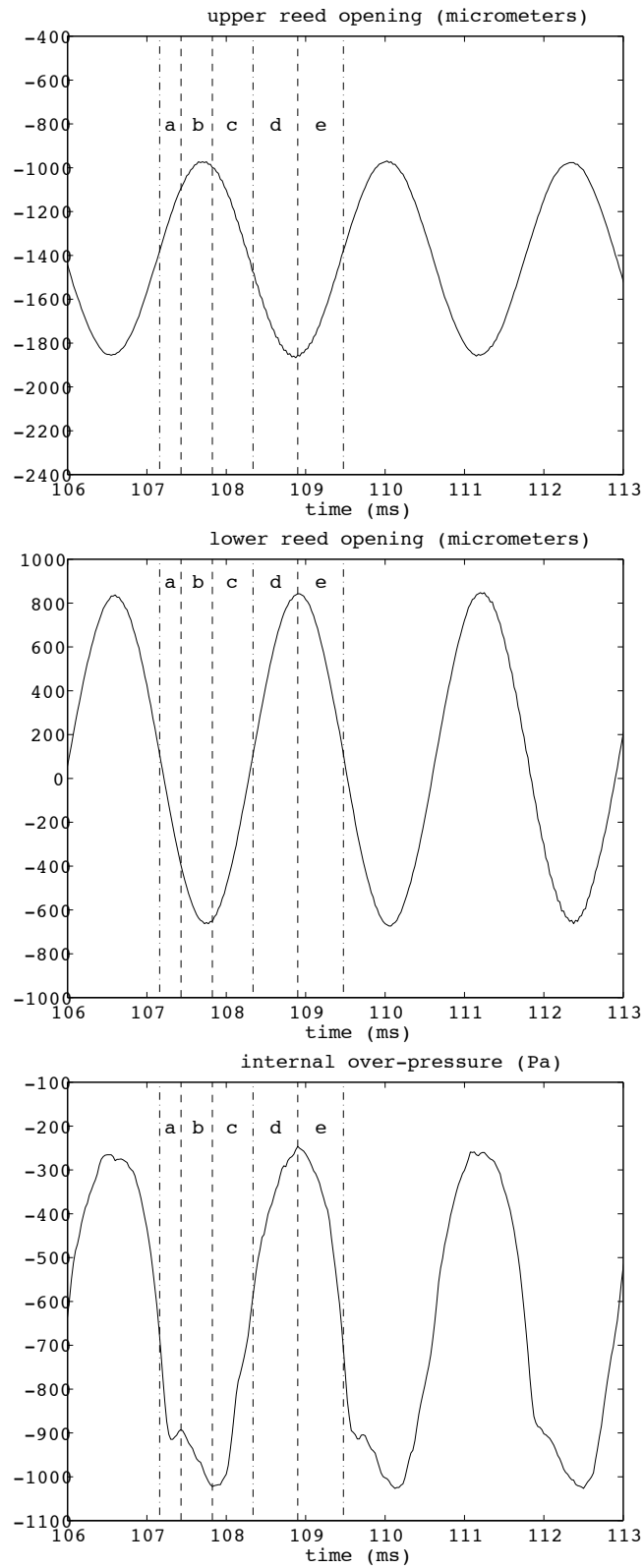


Figure 10: Signals for the upper reed motion, lower reed motion and internal pressure during a drawn bend on channel 4. The dashdot lines indicate the outside closing point of the lower reed while the dashed lines indicate the local extrema of the over-pressure.

Studying the signals for the reed motions and the inner depression (Figure 10), one can distinguish some changes related to the normal mode case. Indeed, as both reeds have large openings, they both have an influence on the inner depression, unlike the normal modes where only one reed is active (the upper reed for a blown note and the lower reed for a drawn note). The changes can be noted in each stage of the waveform cycle for the inner pressure:

- a rise of the depression during stage a, consecutive to the closing of the lower reed, more violent and important than for a normal mode (first peak);
- a second rise of the depression (second peak), during stage b, related to the minimal opening of the upper reed and the presence of the lower reed inside its support (the lower and upper reed openings are not big enough for going through the support);
- the depression decreases while the upper reed opening increases and the lower reed tries to go back outside the support (stage c);
- as the lower reed is outside its support and increases, the upper reed opening increases too, the depression keep on decreasing to its minimum (stage d);
- the depression increases (stage e) with the decrease of both reed openings;
- a new cycle begins.

The risings and fallings of depression are less violent than in the normal mode case (normal blows or draws) which might explain the fact that the power spectrum of the pressure is less rich in the case of the bends than the power spectrum found for the normal modes.

4.3 Overdraw on channel 7

The last kind of mode to be studied is the overnote. We now give the experimental results for an overdraw on channel 7 of harmonica 7b. As the overnotes are very hard to produce and the corresponding reed amplitudes of vibration very strong, we did not succeed in getting the results for both reeds. Indeed, the lower strain gauge died when trying to produce the overnote so we only present the results for the upper reed (Figure 11 and Table 6). Yet, as illustrated by Figure 11, we pointed out that the overnote is easier to produce when the lower reed is blocked [7, 8]. We were not able to verify that the lower reed oscillations are quite small during the overnote.

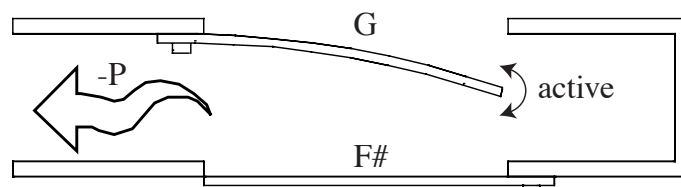


Figure 11: Overdraw configuration according to theory for an overdraw on channel 7.

Table 6: Results during an overdraw on channel 7; harmonica 7b.

reed	f_{play} (Hz)	H_n (μm)	Δh_{n0} (μm)	h_{n0} (μm)
upper 7b	877	590	-241	-1236

On studying the results given by Table 6, we can see that :

- the upper reed auto-oscillates (opening magnitude of 590 μm) and is responsible for the note production;
- the pitch of the played note (877 Hz) is higher than the upper reed eigen frequency (807 Hz);
- the overnote pitch is between the half-tone (845 Hz) and the tone (896 Hz) with respect to the normal blow note (798 Hz);
- as $\Delta h_{10} < 0$ the upper reed is an opening reed;
- the pitch of the played note is higher than the upper reed eigen frequency: the upper reed is a so called *downstream striking reed*.

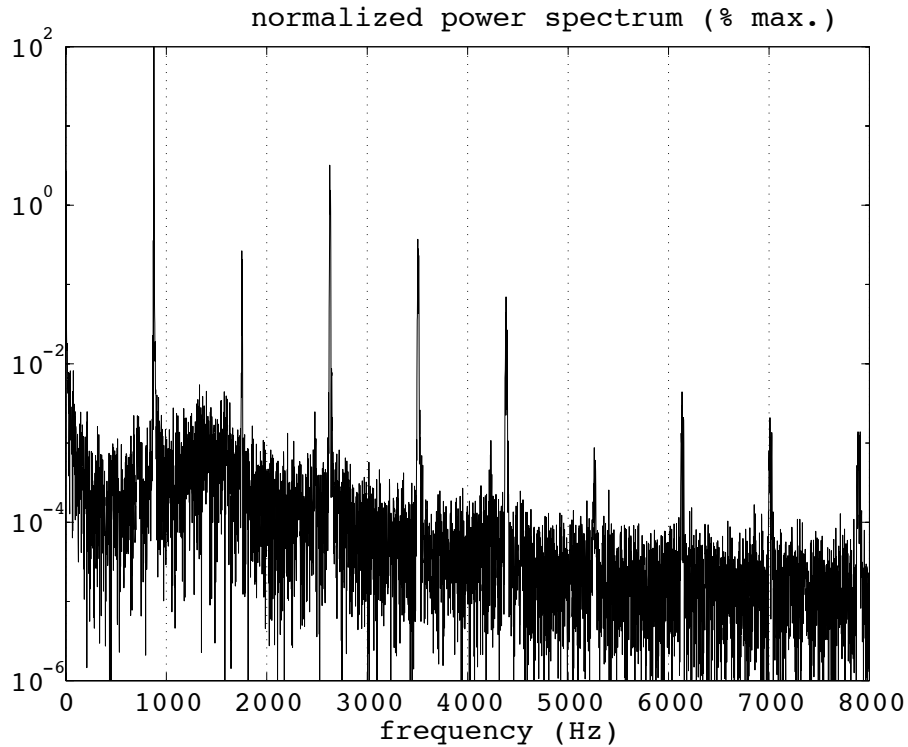


Figure 12: Normalized power spectrum for the internal pressure during an overdraw on channel 7.

As for all the overnotes, producing the overdraw requires a far more important pressure supply, 2482 Pa for the mean depression and a magnitude of 1042 Pa for the oscillations. The related

mean and maximal inner pressure levels, 161.9 dB and 164.9 dB, are also quite important. This is one of the explanations why it is so difficult to produce the overnotes: overnotes are advanced playing modes. Figure 12 illustrates what happens with the power spectrum of the inner depression during the overdaw:

- the fundamental is again the most powerful harmonic;
- the second harmonic is less powerful than the third and fourth ones unlike the normal modes;
- the odd harmonics are more powerful than the following even ones;
- the harmonics above the fourth are very weak, which is also the case for the overnotes, even those produced on harmonicas without strain gauges attached;
- the sixth and seventh harmonics do not have a lot of power.

Although we do not have the information for the lower reed opening, we can explain most of the variations of the inner depression with only the signal for the upper reed opening (Figure 13):

- during stage a, the upper reed is inside the support and we have a corresponding decrease of the depression after a first peak related to the inside closing point of the reed (the upper reed does not go through its support);
- the reed goes back up to its inside closing point (stage b) and the depression increases up to its second peak;
- the reed opening increases up to its inside maximum inside the channel and the depression drops almost to its minimum (stage c);
- during stage d, the reed opening decreases to its closing point while the depression rises gradually;
- a new cycle begins.

One can see that the variations of the inner depression are violent, as in the case of the normal playing modes, and important, as in the case of the bend modes. This might not be surprising because, on one hand, during the overnotes one reed mostly drives the production of the note and, on the other hand, the overnotes are very hard to produce. We must note that the peaks of the depression are not related to the closing points of the upper reed ; we think that the information relative to the lower reed may explain this phase shift.

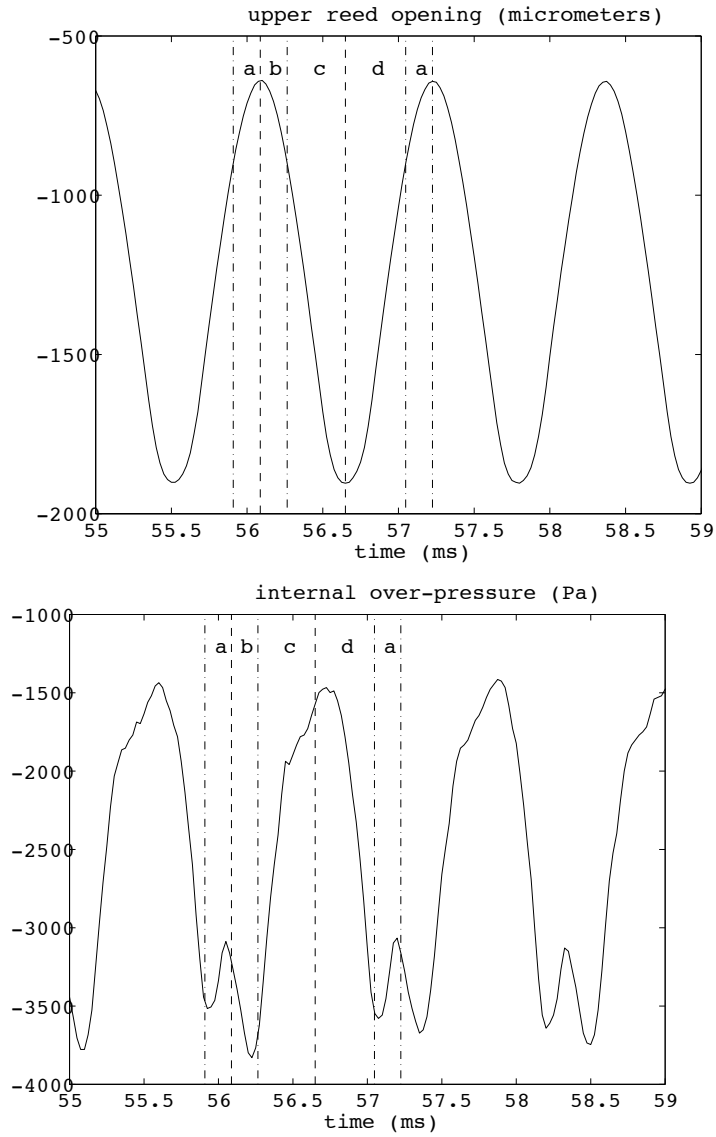


Figure 13: Signals for the upper reed motion and internal pressure during an overdraw on channel 7. The dashdot lines are related to the inside closing point of the upper reed while the dashed lines are related to the extrema of the upper reed opening.

5 Conclusion

Both reed motions and the relative inner pressure were measured, using strain gauges and differential pressure sensors, for the three kinds of playing modes on the diatonic harmonica: normal playing, bends and overnotes.

The measurements pointed out the sinusoidal character of the reed motions when playing, unlike that found qualitatively by Bahnson [8]. Using these observations and the results of the reed calibration, the vibrational and mechanical properties of the equivalent damped oscillators (mass, stiffness, damping, quality factor, plucked eigen frequency and rest offset) were derived for all the tested reeds. The investigations on reed motions showed, for each mode, which reed was active (auto-oscillation) or passive (forced oscillation), the oscillation characteristic of each reed (inwards or outwards striking reeds) and how the pitches of the played notes were related to the plucked natural frequencies of the reeds.

The simultaneous studies of reed motions and inner pressure signals underlined the control of the active reed(s) on the evolutions of the inner pressure. The presence of sharp peaks in the inner pressure waveforms should explain the very rich harmonic content of the sounds produced by diatonic harmonicas. Moreover, as the measured reed motions were always sinusoidal, we think that the non-linear production of harmonics may be mainly explained by the non linear character of the airflow through the reeds.

As some quantitative results for reed motions and behaviour of the inner pressure are now available, we have studied the oral action (fricative vowel) used to produce each kind of mode on diatonic harmonica. Then, after the identification of the oral action, we should be able to derive a model for the diatonic harmonica and use it to establish which are the control parameters for each playing mode.

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